AWS EC2 Disk, CPU & Memory Issues – Interview Notes (Hinglish)

1. CPU Issue in EC2

Symptoms

- Server slow ho jata hai
- Application hang
- High response time
- SSH login slow
- CPU utilization 80–100%

Check Commands (Linux)

```
top
```

```
htop
```

```
mpstat -P ALL 1
```

```
uptime
```

AWS Monitoring

Use AWS CloudWatch:

- CPUUtilization metric
- Set alarms

Common Causes

- Infinite loop process
- Java app high threads
- Too many requests
- Small instance type
- Crypto/mining malware

Fix

Kill unwanted process

```
kill -9 PID
```

Other Fixes

- Restart service
- Scale vertically (t2.micro → t3.large)
- Enable Auto Scaling
- Optimize application

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2. Memory (RAM) Issue in EC2

Symptoms

- Instance freeze
- Out Of Memory (OOM)
- Application crash
- Swap usage high

Check Memory

```
free -h
```

```
vmstat 1
```

```
top
```

```
cat /var/log/messages | grep -i oom
```

Ubuntu:

```
dmesg | grep -i kill
```

Common Causes

- Memory leak
- Too many containers/processes
- Java heap issue
- No swap configured

Fix

Find high memory process

```
ps aux --sort=-%mem | head
```

Add Swap

```
sudo fallocate -l 2G /swapfile
sudo chmod 600 /swapfile
sudo mkswap /swapfile
sudo swapon /swapfile
```

Other Fixes

- Reduce heap usage
- Restart leaking app
- Increase instance RAM

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3. Disk Issue in EC2

Symptoms

- “No space left on device”
- Logs not generating
- Server crash
- High disk latency

Check Disk Usage

```
df -h
```

```
du -sh /*
```

```
lsblk
```

Check IOPS / Disk Performance

```
iostat -xz 1
```

AWS Monitoring

In CloudWatch:

- DiskReadOps
- DiskWriteOps
- BurstBalance (gp2 volumes)
- VolumeQueueLength

Common Causes

- Large logs
- Docker images
- Old backups
- Small EBS volume
- High I/O database workload

Fix

Clean logs

```
truncate -s 0 /var/log/messages
```

Remove old files

```
find / -type f -mtime +7
```

Extend EBS Volume

1. Increase EBS size from AWS Console
2. Extend partition

Example:

```
growpart /dev/xvda 1
```

```
resize2fs /dev/xvda1
```

Important AWS Services Used

Service	Usage
EC2	Virtual server
CloudWatch	Monitoring & alerts
Auto Scaling	Automatic scaling
EBS	Disk storage
CloudTrail	Audit logs

Real Interview Scenario Example

Question:

“CPU suddenly became 100% on production EC2. What will you do?”

Answer:

1. Check CloudWatch metrics
2. SSH into instance
3. Run:

```
top
```

1. Identify high CPU process
2. Check logs
3. Restart service if needed

4. If traffic spike:

5. scale instance

6. add Auto Scaling
7. Create CloudWatch alarm for future

Quick Troubleshooting Flow

CPU

```
High CPU → top → identify process → kill/restart → scale
```

Memory

```
OOM → free -h → check leak → add swap → optimize app
```

Disk

```
Disk full → df -h → clean logs/files → extend EBS
```